

HKL-PDF

What is new about the Software

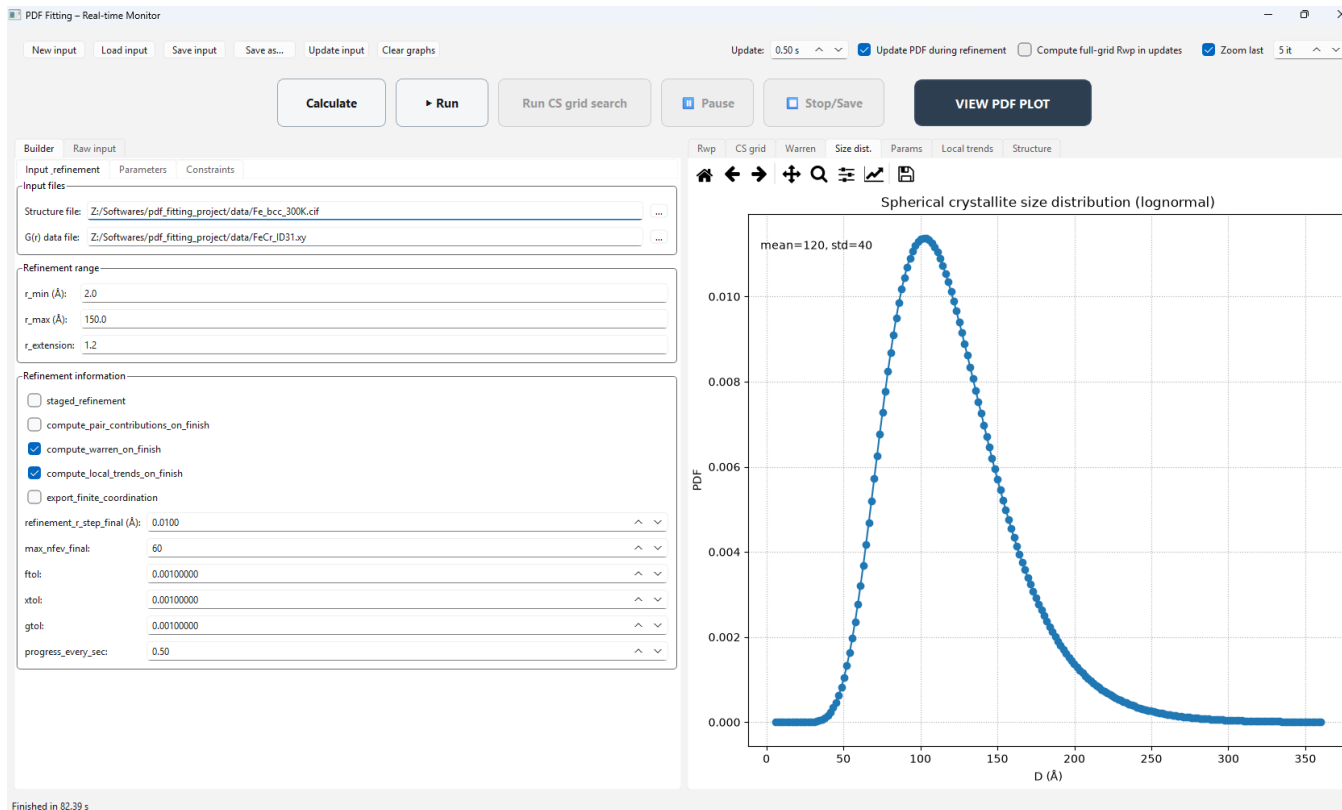
- PDF software to model directional microstructural features such as dislocation microstrain and anisotropic crystallite shapes
- The code relates each pair distance to the HKL (Miller indices), allowing direct comparison with properties obtained in reciprocal space analysis
- Crystallite shape is modelled via:
 - Common Volume Function (CVF) to estimate the shape factor analytically for predefined shapes (like disks, rods, ellipsoids, cubes, etc...)
 - Calculation of the pair distance histogram for random shapes by calculating all the pair distances in the shape, normalized in fractional coordinates (allowing the refinement of lattice parameters, DW factors, etc...). A GUI interface is available for grid search and visualization
- Whole set of visualization tools for Warren Plots, crystallite size distribution, and local dynamics, and refinement monitoring
- It is highly parallelized by CPUs suitable for incorporation into supercomputers. (GPU extension is foreseen).
- It counts with input file structure that does not require GUI interface to run. So one can use the current engine and develop its own GUI interface.

How it works?

- Database creation: given the .cif file, the code calculates all the pair distances up to the maximum pair distance requested by the user. Each pair is associated with an [hkl] direction. Pair-type information, fractional coordinates, coordination numbers, and [hkl] metadata are stored in cache, so the calculations just need to be run once.
- To refine lattice parameters, for example, a simple matrix multiplication using the metric tensor. Broadening parameters such as Debye-Waller parameters (constant in r) and microstrain parameters (that are HKL and r dependent) are assigned for each pair-type and direction and refined
- Crystallite size is modelled by the calculation of the shape factor by CVF (using P. Scardi equations) and histogram calculation for finite shapes.

Visualization tools

Complete user interface for refinement monitoring



Input file structure automatically updated from the GUI builder. So any modification in the GUI is directly translated to code

```
Builder  Raw input
[files]
structure_file = Z:/Softwares/pdf_fitting_project/data/Fe_bcc_300K.cif
gr_data_file = Z:/Softwares/pdf_fitting_project/data/FeCr_ID31.xy

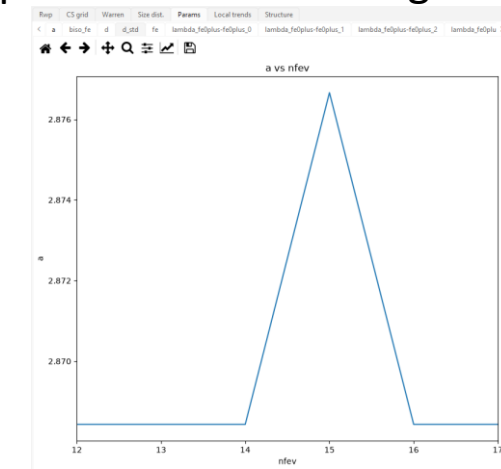
[range]
r_min = 2.0
r_max = 150.0

[pair_generation]
r_extension = 1.2

[refinement]
staged_refinement = false
microstrain_model = isotropic
saxs_enabled = false
saxs_model = sphere
saxs_apply_qdamp = false
saxs_apply_qmax = true
compute_pair_contributions_on_finish = false
compute_warren_on_finish = true
compute_local_trends_on_finish = true
export_finite_coordination = false
refinement_r_step_final = 0.01
max_nfev_final = 60
ftol = 0.001
xtol = 0.001
gtol = 0.001
progress_every_sec = 0.5

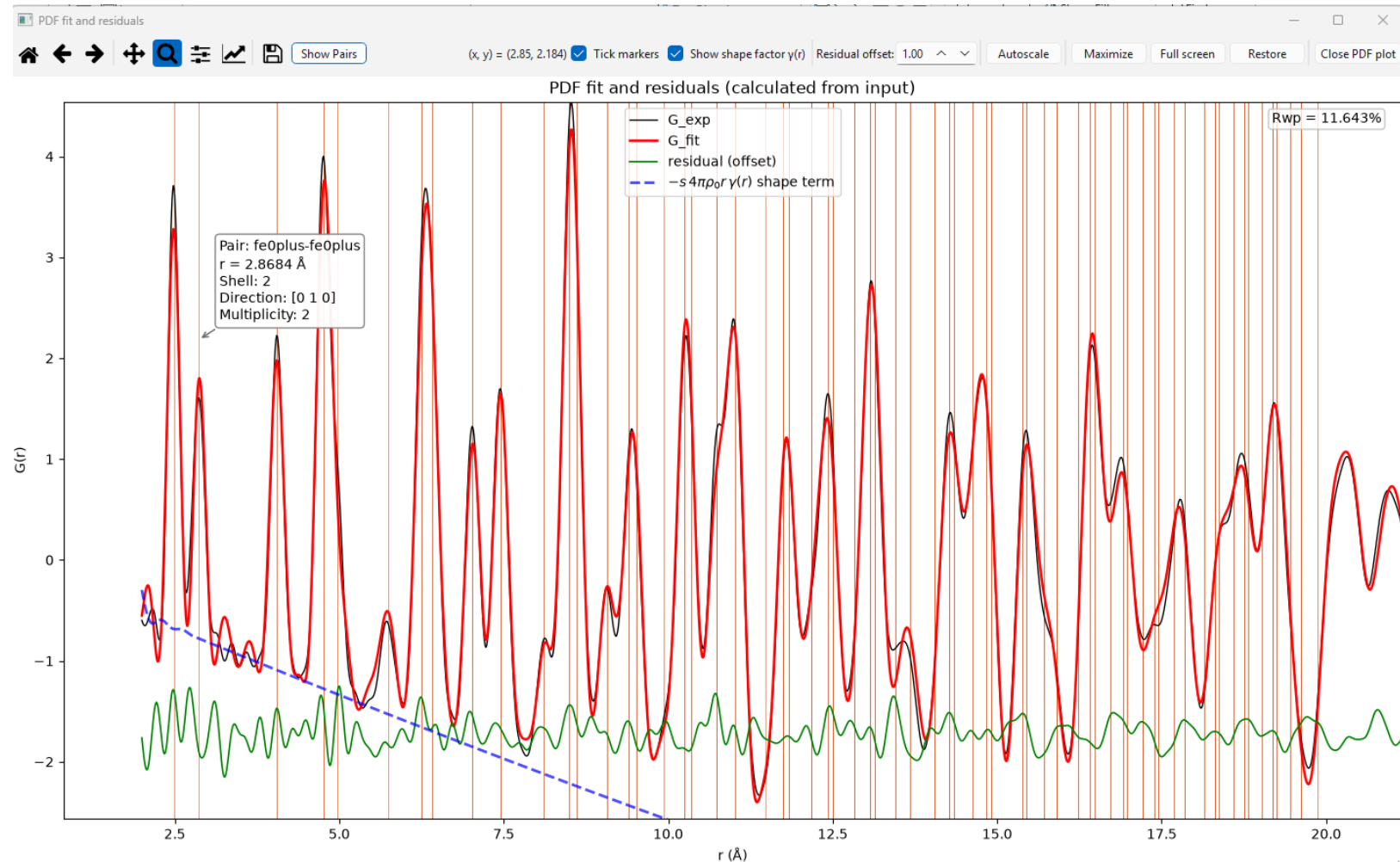
[initial_values]
scale = 0.2500986518331641
qdamp = 0.011485099495637478
delta_broad = 0.0014090180673610453
qmax = 20.0
qmax_zeros = 5.0
d = 84.21033439198368
d_std = 69.33152658229807
a = 2.8683573541584586
b = 2.8683573541584586
c = 2.8683573541584586
alpha = 90.0
beta = 90.0
gamma = 90.0
x_fe1 = 0.0
y_fe1 = 0.0
z_fe1 = 0.0
occ_fe1 = 1.0
biso_fe1 = 0.49416499422968185
```

Real time parameter monitoring during refinement



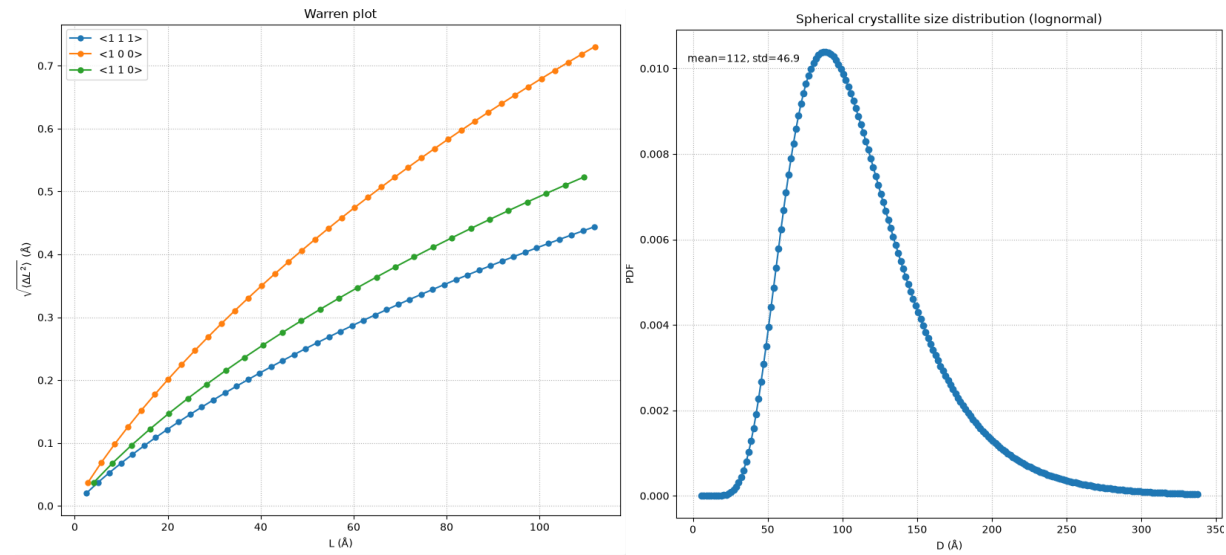
Visualization tools

Fitting visualization tool including tick markers for each pair with information on i) pair-type, ii) distance, iii) shell, iv) direction, v) multiplicity. Full customizable colors are available.

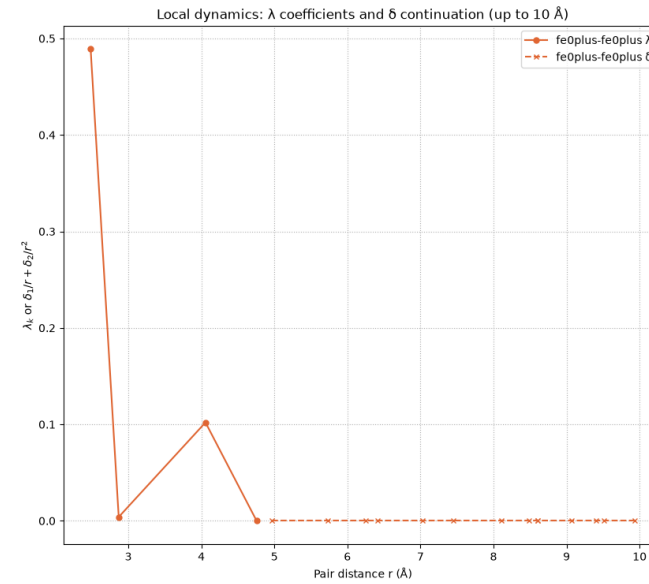


Visualization tools

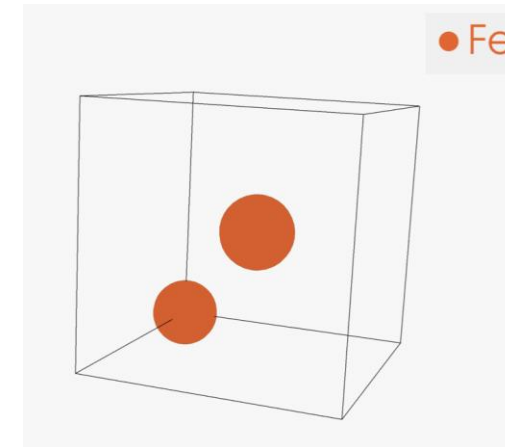
Full microstructural analysis



Local structure visualization tools

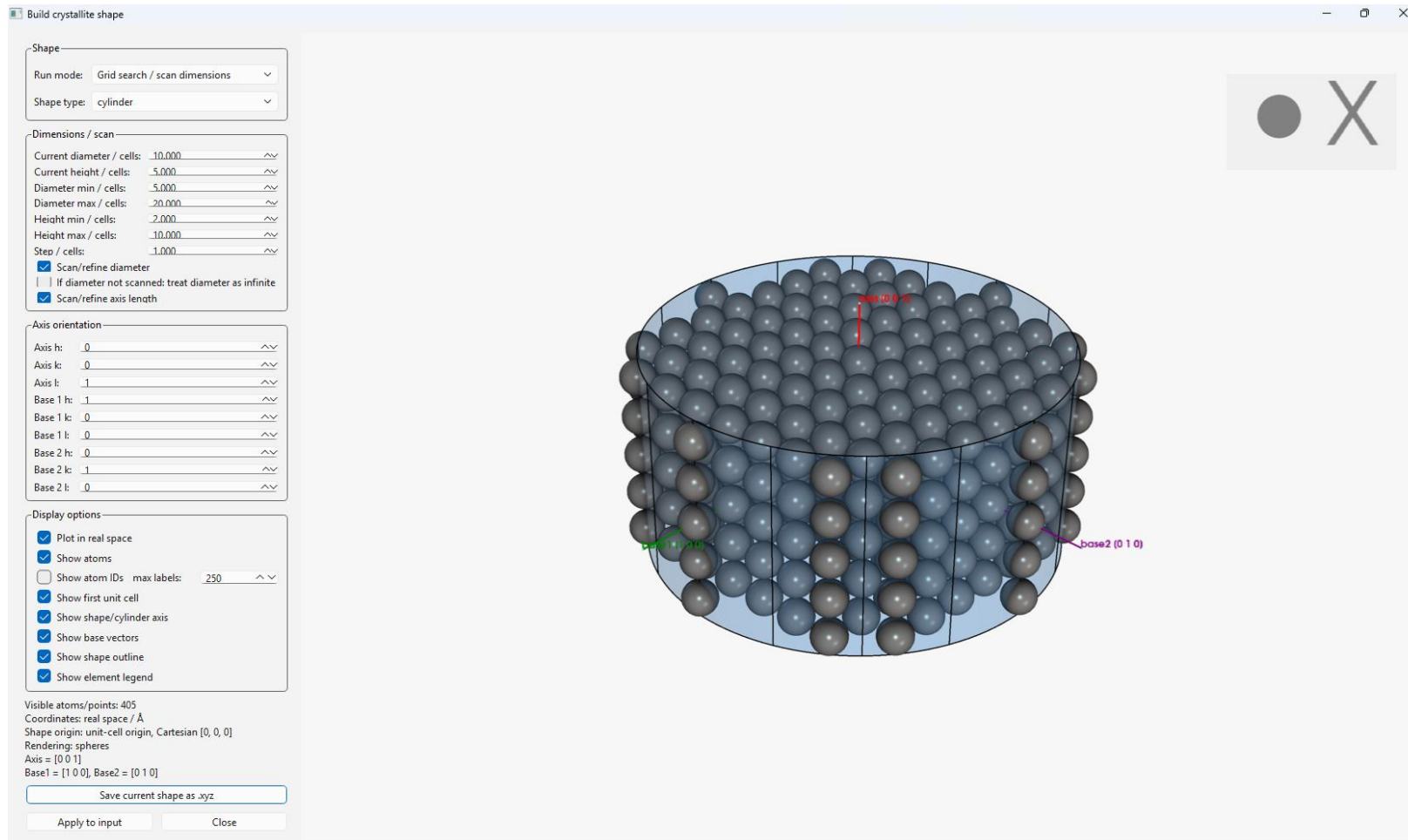


Assymmetric unit



Visualization tool

Finite crystallite shape builder



SAXS contribution

☒ SAXS / missing-low-Q shape belly

SAXS belly model: sphere

☐ Apply qdamp to SAXS belly

☒ Apply qmax termination to SAXS belly

These parameters control only the smooth negative shape/SAXS belly term, not the PDF peak envelope.

	Parameter	Value	Min	Max	Refine?
1	saxs_scale	1.0			☐
2	saxs_diameter	0.0			☐
3	saxs_diameter_std	0.0			☐
	...	...			☐

How to install and run

- Follow GitHub instructions:
<https://github.com/MalaguttiMarcelo/HKL-PDF.git>
- Run the GUI HKL-PDF.exe
- Documentation is available in the docs folder containing all the equations.
- Follow the three examples:
 1. Ball-milled FeCr
 2. Ball-milled CaCO₃
 3. Simulated Pd disks

Future developments

1. Multiphase analysis
2. Atomic-position refinement (XYZ)
3. Automated calculation of shape functions—or common-volume functions—for different shapes
4. A user-friendly module for generating stacking faults (needed for the complex organic materials developed by Bettina).